

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	IBM Bob AI Super Sprint: Python-Controlled Racing Game Challenge
<i>Company:</i>	IBM
Contact Details	
<i>Lead Contact:</i>	John McNamara
<i>Lead E-mail:</i>	J0nnymac@uk.ibm.com
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
<i>Additional Contact:</i>	Additional contact name
<i>E-mail:</i>	Additional contact e-mail
Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Artificial intelligence for games; reinforcement learning and search; autonomous control; software engineering; human-computer interaction; gamified learning; educational technology; digital skills development. Artificial intelligence for games; reinforcement learning and search; autonomous control; software engineering; human-computer interaction; gamified learning; educational technology; digital skills development.
Project Background Information	
is project asks students to build a retro-inspired, top-down racing game in the spirit of the Super Sprint reference video provided by IBM. The aim is not to copy the original game or its assets, but to recreate the core learning pattern: short races, clear tracks, fast feedback, simple controls and measurable improvement. The distinctive educational feature is that every car is governed by a Python strategy script. Students are given a ready-made Python template containing functions for throttle, braking, steering, route choice, overtaking, collision avoidance, power-up collection and weapon deployment. Their challenge is to improve that template so the car becomes faster, safer and more competitive. IBM Bob should be used as the learning accelerator. Players use IBM Bob to understand the starter template, generate or improve Python logic, explain errors, compare approaches, refactor code and document the reasoning behind their racing strategy. The resulting activity can be run as a hackathon,	

¹ An overview of different research areas in the school of computer science can be found [here](#).

classroom challenge or university showcase in which teams compete on lap time, race wins, code quality, learning reflection and effective use of IBM Bob.

Requirements

The group must provide:

- A playable top-down racing game prototype inspired by the Super Sprint format, using original or openly licensed assets rather than copied game material.
- A Python agent interface that controls each AI car, including acceleration, braking, steering, target racing line, overtaking behaviour and obstacle avoidance.
- A ready-made Python starter template with clearly labelled TODO sections so participants can improve the AI without needing to understand the whole game engine first.
- At least one baseline bot and two or more enhanced bots that demonstrate measurable performance improvement through better racing line, speed management, overtaking or power-up strategy.
- Power-ups and optional weapons such as boost, shield, repair, oil slick or temporary speed reduction, with Python logic determining when to collect and deploy them.
- A scoring system suitable for a hackathon: best lap time, race result, consistency, safe driving, successful power-up use, code quality and learning reflection.
- IBM Bob usage guidance inside the project, such as prompt examples, debugging suggestions and explainability notes that help students improve the Python template responsibly.
- A short technical report explaining the game architecture, Python API, AI decision logic, testing method, limitations and recommended future enhancements.
- A final demonstration, ideally including a short video or live race, showing human-readable code changes and the performance impact on the track.

Skills

Highly desirable:

- Strong Python programming fundamentals
- Interest in AI/game AI, including search, heuristics, reinforcement learning or rule-based agents
- Game development experience using Pygame, Godot, Unity or similar
- Ability to design clean APIs and test autonomous behaviours
- Git/GitHub collaboration and clear technical documentation

Desirable:

- Experience using IBM Bob or other AI-assisted development tools
- Basic physics, geometry, vectors or path-following algorithms
- UI/UX and visual design for accessible learning experiences
- Hackathon/event design and scoreboard mechanics
- Interest in IBM Granite, watsonx or IBM SkillsBuild learning pathways

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.